

Assignment 04: Advanced Topics in AI

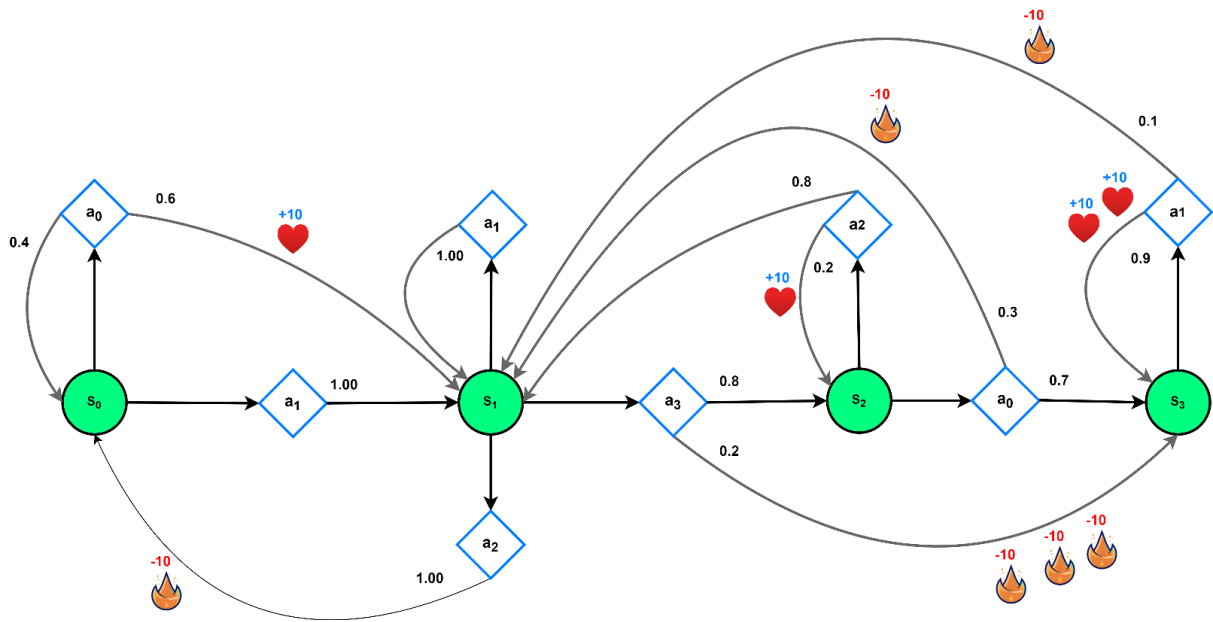
Q1: Bellman Optimality Equation [20 pts].

As we studied in-Class, for the MDP provided below, write a program to find the optimal action for each state.

Deliverables:

- Report with code snapshots.
- Code File.

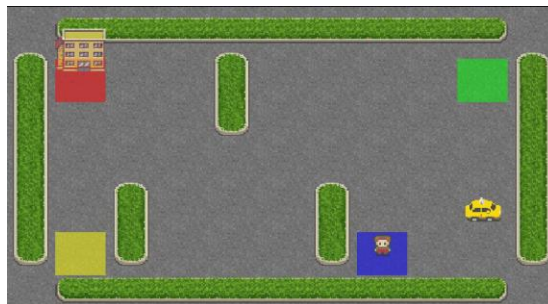
Notes: Codes should be neat, well-commented, and properly structured.



Reference: [18. Reinforcement Learning | Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 3rd Edition \(oreilly.com\)](#)

Q2: Reinforcement Learning [80 pts].

You will work with the [Taxi environment](#) and implement Q-Learning (OpenAI, n.d.). The way you interact with the environment will be very similar to the cartpole example described in class. Therefore, most of the pseudo code we discussed is directly applicable.



The environment has **6 discrete actions**, **500 discrete states**, and the following rewards:

- -1 per step unless other reward is triggered.
- +20 delivering passenger.
- -10 executing “pickiup” and “drop-off” actions illegally.

Action space

- 0: Move south (down)
- 1: Move north (up)
- 2: Move east (right)
- 3: Move west (left)
- 4: Pickup passenger
- 5: Drop off passenger

Observation Space

There are 500 discrete states since there are 25 taxi positions, 5 possible locations of the passenger (including the case when the passenger is in the taxi). and 4 destination locations. Destinations on the map are represented with the first letter of the color.

Passenger locations:

- 0: Red
- 1: Green
- 2: Yellow
- 3: Blue
- 4: In taxi

Destinations:

- 0: Red
- 1: Green
- 2: Yellow
- 3: Blue

An observation is returned as an `int()` that encodes the corresponding state, calculated by $((taxi_row * 5 + taxi_col) * 5 + passenger_location) * 4 + destination$.

Problem Statement

Implement the Q-Learning algorithm on the Taxi environment from OpenAI Gym. Train an agent to efficiently navigate and pick up passengers. Use the following hyperparameters:

- Learning Rate α : 0.1
- Exploration Factor ϵ : 0.1
- Discount Factor γ : 0.9

Tasks

- Implement the Q-Learning algorithm and train an agent on the Taxi environment.
- Report the following metrics after training:
 1. Total episodes.
 2. Total steps taken per episode.
 3. Return per episode.
- Make a deliberate change to the following parameters, **SEPARATELY**, and use each value once.
 - Learning Rate $\alpha = [0.01, 0.001, 0.2]$
 - Exploration Factor $\epsilon = [0.2, 0.3]$

Deliverables

- Python code implementing Q-Learning and running it for the different hyperparameters. [20].
- A report (in PDF) containing the metrics, observations, and comments on the parameter change. Make sure you include appropriate plots for the previously stated metrics to support your arguments. A well-documented Jupiter Notebook submission is accepted too [30].
- Based on your findings, choose what you think would be the best combination of learning rate and exploration factor and re-run the training. Report and comment on the differences observed. [30]

References

OpenAI. (n.d.). *Introduction to Q-learning with OpenAI Gym*. Retrieved from Open AI:
https://gymnasium.farama.org/environments/toy_text/taxi/#taxi